Unit 1 — General & Systematic Veterinary Bacteriology

All 34 syllabus topics compressed to six working sheets. Built so that one uninterrupted 30-minute pass gives you the diagram, the differentiating table, the one-liner and the answer skeleton for every examinable item in Paper I, Unit 1.

PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper I · Units 1, 2, 3 Theory 100 · Practical 60	34 of 34 covered	30 min full pass 12 min sheets 5–6 only	B.V.Sc & A.H. II Yr VCI MSVE · 3+2=5

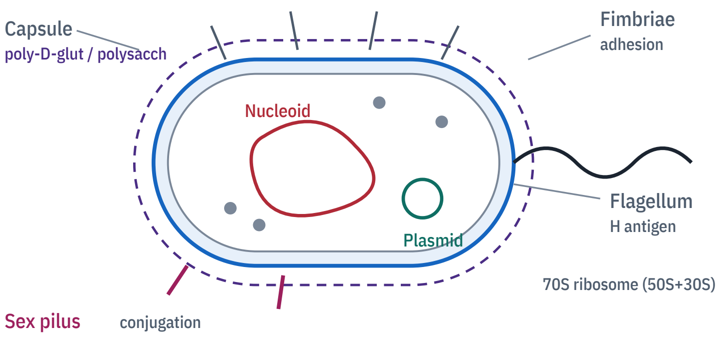
Sheet	What it covers	Syllabus topics folded in
01	The bacterial cell — structure, staining, growth, microscopy	t01–t07
02	Infection, virulence, toxins, genetics, antibiotic resistance	t08–t12
03	Gram-positive cocci & non-sporing rods	t13–t16
04	Spore-formers, acid-fast & filamentous bacteria	t17–t20
05	Gram-negative rods — the whole enteric & respiratory block	t21–t28
06	Anaerobes, spirochaetes, cell-wall-deficient & obligate intracellular	t29–t34

General / structural · Gram-positive · Gram-negative · Acid-fast · Anaerobe / spore · Exam trap & mnemonic

01 The Bacterial Cell — Structure, Stains, Growth

Cell architecture

MECHANISM



- **Cell wall** = peptidoglycan (murein): NAM–NAG backbone, cross-linked by transpeptidase — the **penicillin target**. Absent in *Mycoplasma*.
- **Gram +ve**: 20–80 nm thick peptidoglycan (~50–90% of wall), **teichoic + lipoteichoic acid**, no outer membrane, no periplasm.
- **Gram −ve**: 2–7 nm peptidoglycan, **outer membrane with LPS**, periplasmic space, porins, Braun's lipoprotein.
- **LPS = Lipid A (toxic, endotoxin) + core + O-side chain (O antigen)**.
- Mesosome = infolding of plasma membrane; site of septum formation & DNA segregation.
- Spore layers outward: core → cortex (**Ca-dipicolinate**) → spore coat → exosporium.

Gram-positive vs Gram-negative

TABLE

Feature	Gram +ve	Gram −ve
Final colour	Violet	Pink / red
Peptidoglycan	Thick, multilayer	Thin, single layer
Teichoic acid	Present	Absent
Outer memb. / LPS	Absent	Present
Periplasm	Absent	Present
Toxin	Mostly exotoxin	Endotoxin ± exotoxin
Lysozyme	Sensitive	Resistant (OM barrier)
Penicillin	Sensitive	Relatively resistant
Wall-less form	Protoplast	Spheroplast
Flagellar rings	2 (S, M)	4 (L, P, S, M)

Trap: Gram reaction is a *wall* property, not a taxonomic one — old, autolysed or antibiotic-treated Gram +ve cells stain Gram-variable. *Mycobacterium* is technically Gram +ve but does not take Gram stain.

Staining — what each step actually does

MECHANISM

1 · Crystal violet — 60 s — primary stain
2 · Gram's iodine — mordant → CV-I complex
3 · Decolouriser — acetone-alcohol, 5–10 s
4 · Safranin — counterstain, 30 s

Gram POSITIVE
Thick peptidoglycan dehydrates, pores shut → CV-I trapped → stays VIOLET

Gram NEGATIVE
Alcohol dissolves the lipid-rich outer membrane → CV-I washes out → takes PINK

Single most common mistake: over-decolourisation. Beyond ~10 s even true Gram +ve cells go pink. Decolouriser is the *only* timed-to-the-second step.

Stains you must be able to name

RAPID-FIRE

Simple	Methylene blue / basic fuchsin — shape & arrangement only
Negative	India ink, nigrosin — *Cryptococcus* capsule; background stained, cell clear
Gram (1884)	Christian Gram — differential; violet vs pink
Ziehl-Neelsen	Carbol fuchsin + heat, 20% H_2SO_4 + alcohol, methylene blue → AFB red on blue
Modified ZN	0.5–1% H_2SO_4 only — *Brucella*, *Nocardia*, *Coxiella*, *Chlamydia* (partially acid-fast)
Capsule	McFadyean's (polychrome methylene blue) — blue bacilli, pink capsule; Anthony's ($CuSO_4$)
Spore	Schaeffer-Fulton — malachite green (heat) + safranin → green spore, red vegetative cell
Flagella	Leifson / Ryu — mordant-thickened; Fontana for spirochaetes (silver impregnation)
Metachromatic granules	Albert's, Neisser's, Loeffler's methylene blue — *Corynebacterium* (Babes-Ernst granules)
Giemsa	Blood parasites, *Anaplasma*, *Chlamydia* inclusions, *Haemobartonella*

Microscopy & micrometry

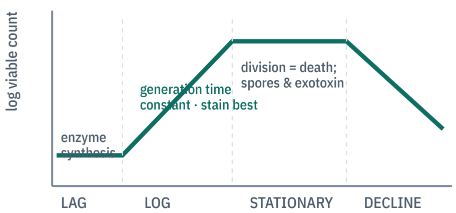
RAPID-FIRE

Resolving power	RP = 0.61λ / NA
Light micro. limit	≈ 0.2 µm; useful mag ≈ 1000–1500×
Oil immersion	Cedar wood oil, RI 1.515 = glass; raises NA to 1.25–1.4
Dark field	Live, unstained — *Leptospira*, *Treponema*
Phase contrast	Live unstained cells, internal detail
Fluorescence	Auramine-rhodamine for AFB; FAT for rabies
Electron micro.	TEM resolves ≈ 0.2 nm — only way to see virions
Micrometry	Stage micrometer calibrates the ocular micrometer; 1 stage div = 10 µm

Calibration 1 ocular div (µm) = (stage divs coinciding × 10) ÷ ocular divs coinciding. Recalibrate for *every* objective.

Growth curve & requirements

MECHANISM



- **Generation time**: *E. coli* 20 min · *M. tuberculosis* 18–24 h · *M. leprae* ~13 d.
- Temperature: psychrophile <20 · **mesophile 25–40 (all pathogens)** · thermophile 55–80.
- pH: most 7.2–7.4; *Vibrio* alkaline 8.5–9.0; fungi acidic 5.6.
- O_2: obligate aerobe · facultative anaerobe · **microaerophile (*Campylobacter* 5% O_2)** · obligate anaerobe · capnophile (5–10% CO_2 — *Brucella abortus*).

Media — the exam list

RAPID-FIRE

Basal	Nutrient broth/agar, peptone water
Enriched	Blood agar, chocolate agar, serum agar
Enrichment (liquid)	Selenite F, tetrathionate, alkaline peptone water
Selective	MacConkey, DCA, TCBS, Lowenstein-Jensen, SDA
Differential	MacConkey (lactose), blood agar (haemolysis), EMB
Indicator	Wilson-Blair (black colonies of *Salmonella*)
Transport	Cary-Blair, Stuart's, Amies
Anaerobic	Robertson's cooked meat, thioglycollate, Brewer's
Sugar-free	For toxin studies of *C. diphtheriae*

Anaerobiosis McIntosh–Fildes jar · GasPak · Robertson's meat (unsaturated fat + glutathione) · **Methylene blue indicator: blue = O_2 present, colourless = anaerobic.**

Normal, opportunistic & saprophytic flora

TABLE

Site	Predominant normal flora	Note
Skin	*Staphylococcus* (coag −ve), *Micrococcus*, *Corynebacterium*	Source of surgical-site infection
Rumen	Cellulolytic bacteria, protozoa, anaerobic fungi	**Symbiotic — VFA supply**
Intestine	*E. coli*, *Lactobacillus*, *Bacteroides*, *Clostridium*	Vitamin K & B synthesis; colonisation resistance
Upper resp.	*Pasteurella*, *Mannheimia*, streptococci	Stress → shipping fever
Teat canal	CNS, *Corynebacterium bovis*	Keratin plug is first defence
Sterile sites	Blood, CSF, muscle, internal organs, urinary bladder, uterus of healthy animal	

Define cleanly: Commensal = benefits, host unharmed. **Opportunist** = normal flora turning pathogen when host defence falls (immunosuppression, catheter, prolonged antibiotics). **Saprophyte** = lives on dead organic matter, rarely invades.

History — dates that get asked

RAPID-FIRE

Leeuwenhoek 1676	First saw "animalcules"	Jenner 1796	Cowpox → smallpox vaccine
Pasteur	Germ theory, attenuation, anthrax & rabies vaccine, pasteurisation	Koch 1876/1882	*B. anthracis*; tubercle bacillus; solid media, postulates
Lister 1867	Antisepsis — carbolic acid	Ehrlich	Salvarsan; side-chain theory
Fleming 1928	Penicillin from *P. notatum*	Iwanowski 1892	TMV — birth of virology
Gram 1884	Differential staining	Petri 1887	Petri dish

Koch's postulates — write all four

1. Organism present in *every* case, absent from healthy animals.
2. Isolated and grown in **pure culture**.
3. Pure culture reproduces the same disease in a susceptible host.
4. Re-isolated from the experimentally infected host.

Exceptions to quote: *M. leprae* & *T. pallidum* (not cultivable), viruses/prions, carriers, polymicrobial & toxin-mediated disease. Modern answer: **molecular Koch's postulates** (gene knockout abolishes virulence, restoration returns it).

Classification & nomenclature

RAPID-FIRE

Reference	Bergey's Manual of Systematic Bacteriology — the standard
Binomial rule	Genus capitalised, species lower case, both italic/underlined: *Staphylococcus aureus*; abbreviate *S. aureus* after first use

Sheet 01 · 60-second self-test

RECALL

Prompt	Answer
Mordant of Gram stain	Gram's iodine
Chemical unique to bacterial spore	Calcium dipicolinate

Hierarchy	Domain→Phylum→Class→Order→Family→Genus→Species
Species (bacteria)	Strains with ≥70% DNA–DNA hybridisation, <5 °C ΔTm
Gold standard now	**16S rRNA gene sequencing** (~1500 bp; >97% identity = same species)
Sub-species terms	Serovar (antigen) · biovar (biochemical) · pathovar · phagovar · biotype
Type strain	Nomenclatural reference; deposited (ATCC, NCTC, MTCC)
Other schemes	Numerical taxonomy (Adansonian), chemotaxonomy, MLST, MALDI-TOF MS

Prompt	Answer
Toxic moiety of endotoxin	Lipid A
Stain for anthrax capsule	McFadyean's
Acid used in modified ZN	0.5–1% H_2SO_4
Enzyme cross-linking peptidoglycan	Transpeptidase (PBP)
Phase with maximum exotoxin/spores	Stationary phase
Ribosome of bacteria	70S = 50S + 30S
Medium for *Vibrio*	TCBS / alkaline peptone water
Anaerobic indicator dye	Methylene blue

02 Infection, Virulence, Toxins, Genetics & Resistance

t08–t12

Infection — the vocabulary that carries marks

RAPID-FIRE

Infection	Entry, multiplication & establishment of an agent in host	Contamination	Presence on surface, no multiplication
Primary	First infection in a fresh host	Secondary	New agent on a weakened host
Focal	Confined to one site	Cross	New agent acquired from another patient/hospital
Nosocomial	Hospital-acquired	Iatrogenic	Caused by a procedure/physician
Latent	Agent persists silently, reactivates	Sub-clinical	Infected, no signs, still sheds
Zoonosis	Animal → human (anthrax, brucellosis, TB, leptospirosis)	Carrier	Healthy, incubatory, convalescent, chronic, contact

Route	How it happens	Example
Ingestion	Contaminated feed/water/milk	Salmonellosis, Johne's
Inhalation	Aerosol, droplet nuclei	Tuberculosis, pasteurellosis
Contact	Direct / fomite / venereal	Dermatophilosis, campylobacteriosis
Vector	Biological or mechanical arthropod	Anaplasmosis, HS spread by flies
Vertical	Transplacental, transovarian, milk	Brucellosis, *S. pullorum*
Iatrogenic	Needles, instruments, surgery	Tetanus, abscess

Pathogenicity → disease

MECHANISM

1 · ADHESION & COLONISATION
Fimbriae (K88/F4, K99/F5, 987P), LTA, biofilm

2 · INVASION
Invasins, T3SS injectisome, hyaluronidase, collagenase

3 · EVASION OF DEFENCE
Capsule (antiphagocytic), protein A, coagulase,
IgA protease, catalase, antigenic variation, intracellular surviv

4 · IRON ACQUISITION
Siderophores — enterobactin, aerobactin; Tf/Lf receptors

5 · DAMAGE
Exotoxin, endotoxin, enzymes, immunopathology

DISEASE

Define **Pathogenicity** = *qualitative* ability of a species to cause disease. **Virulence** = *quantitative* degree of pathogenicity, measured as **LD_{50} / ID_{50}**. Lower LD_{50} = more virulent.

Exotoxin vs Endotoxin

TABLE

Feature	Exotoxin	Endotoxin
Nature	Protein (polypeptide)	LPS — Lipid A
Source	Mostly Gram +ve, some −ve; secreted	Gram −ve outer membrane; released on lysis
Heat	Labile (60 °C/30 min) — **except staph enterotoxin**	Stable (121 °C, 1 h)
Potency	Highly potent (botulinum most potent known)	Weak, needs large dose
Specificity	Specific receptor & target tissue	Non-specific — fever, DIC, shock
Antigenicity	Strong; neutralised by antitoxin	Poor; antibody not protective
Toxoid	**Yes** (formalin 0.3–0.4%, 37 °C)	Cannot be toxoided
Genes	Plasmid / lysogenic phage often	Chromosomal
Detection	Neutralisation, ELISA, mouse test	**LAL (Limulus amoebocyte lysate)**

Also define: Bacteraemia = organisms in blood, transient, no multiplication. **Septicaemia** = multiplication in blood + systemic signs. **Toxaemia** = toxin in blood without the organism (tetanus, enterotoxaemia). **Pyaemia** = pus-forming organisms in blood → metastatic abscesses.

Named toxins to memorise

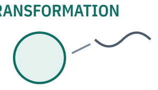
RAPID-FIRE

A-B model	B = binding subunit, A = active enzymatic subunit
Diphtheria & *Pseudomonas* ExoA	ADP-ribosylate EF-2 → protein synthesis stops
Cholera / *E. coli* LT	ADP-ribosylate Gs → ↑cAMP → secretory diarrhoea
E. coli ST	↑cGMP via guanylate cyclase C
Tetanospasmin	Cleaves synaptobrevin → blocks glycine/GABA → **spastic** paralysis
Botulinum	Cleaves SNARE → blocks ACh release → **flaccid** paralysis
Anthrax	PA + EF (adenylate cyclase, oedema) + LF (protease, MAPKK)
α-toxin *C. perfringens*	Lecithinase (phospholipase C) — Nagler reaction
ε-toxin *C. perfringens* D	Pulpy kidney; permease, ↑vascular permeability
Staph α / PVL / TSST-1	Pore-forming / leucocidin / superantigen
Streptolysin O & S	O = oxygen-labile, antigenic (ASO titre); S = stable, non-antigenic
Superantigen	Bridges MHC-II to TCR Vβ outside groove → mass T-cell activation, cytokine storm

Gene transfer — the three routes

MECHANISM

TRANSFORMATION



Naked DNA taken up by *competent* cell · Griffith 1928
Avery, MacLeod & McCarty 1944

TRANSDUCTION

Bacteriophage carries DNA

Generalised — lytic, any gene **Specialised** — lysogenic, adjacent genes Zinder & Lederberg 1952 CONJUGATION Sex pilus, cell contact · $F^+ \rightarrow F^-$ Hfr = integrated F · Lederberg & Tatum 1946

Transposons ("jumping genes") move within/between replicons and carry resistance cassettes. **Integrons** capture gene cassettes by site-specific recombination — the engine of multi-drug R plasmids.

Mutation & plasmids

RAPID-FIRE

Point mutation	Substitution — missense, nonsense, silent
Frameshift	Insertion / deletion (not ×3)
Spontaneous rate	10^{-6} – 10^{-9} per cell per division
Mutagens	UV (thymine dimers), alkylating agents, base analogues, acridine dyes
Ames test	Mutagenicity screen using *S. typhimurium* his^-
F plasmid	Fertility — conjugation
R plasmid	RTF + r-determinant; multidrug resistance
Col plasmid	Bacteriocins (colicin)
Virulence plasmid	pXO1/pXO2 anthrax; ETEC enterotoxin; *Yersinia* pYV
Episome	Plasmid able to integrate into chromosome
Curing	Loss of plasmid — acridine orange, ethidium bromide, heat

Antibiotic resistance

MECHANISM

Mechanism	Example
Enzymatic destruction	β-lactamase, ESBL, **AmpC**, carbapenemase (NDM-1); acetyl/adenyl transferase for aminoglycosides
Target alteration	**PBP2a (*mecA*) = MRSA**; ribosomal methylation (*erm*); DNA gyrase mutation → quinolone R
Reduced uptake	Porin loss (*Pseudomonas*)
Efflux pumps	Tetracycline (*tet*), multidrug RND pumps
Bypass pathway	Sulphonamide resistance via exogenous folate
Biofilm / persisters	Phenotypic tolerance, not genetic

Vet relevance Sub-therapeutic in-feed antibiotics & poor withdrawal-period compliance drive AMR into the food chain. **MRSA, ESBL *E. coli*, MDR *Salmonella*** are the named "One Health" examples. India: NAP-AMR; ban on colistin as growth promoter (2019).

Long-answer skeleton — "Virulence determinants of bacteria" SKELETON

Ten-line frame that fits any 10–15 mark question

1. **Define** pathogenicity vs virulence; state LD_{50}/ID_{50}.
2. **Adhesins** — fimbrial (K88, K99, 987P, F41) and afimbrial; tissue tropism.
3. **Invasion factors** — invasins, T3SS, spreading enzymes (hyaluronidase = "spreading factor", collagenase, fibrinolysin).
4. **Antiphagocytic** — capsule, M protein, protein A, coagulase, leucocidin.
5. **Intracellular survival** — inhibition of phagolysosome fusion (*M. tuberculosis*), escape to cytosol (*Listeria*, listeriolysin O), catalase/SOD.
6. **Iron piracy** — siderophores.
7. **Toxins** — exotoxin classes (neuro-, entero-, cyto-, superantigen) + endotoxin.
8. **Antigenic variation & molecular mimicry**.
9. **Biofilm** and quorum sensing.
10. **Genetic basis** — pathogenicity islands, plasmids, lysogenic conversion; close with one Indian field example.

Sheet 02 · 60-second self-test RECALL

Prompt	Answer
Most potent bacterial toxin	Botulinum
Toxin causing spastic paralysis	Tetanospasmin
Test to detect endotoxin	LAL test
Gene for methicillin resistance	*mecA* → PBP2a
Transduction discovered by	Zinder & Lederberg, 1952
Transformation discovered by	Griffith, 1928
Plasmid-free strain produced by	Curing (acridine orange)
Enzyme called "spreading factor"	Hyaluronidase
Toxin ADP-ribosylating EF-2	Diphtheria toxin
Iron-chelating virulence molecule	Siderophore

03 Gram-Positive Cocci & Non-Sporing Rods

t13–t16

Sorting the Gram-positives at the bench MECHANISM

Gram-positive isolate

COCCI — CATALASE

Catalase +ve
Clusters · *Staphylococcus*
Coagulase + → *S. aureus*
Coagulase – → CNS (*S. epidermidis*)

Catalase –ve
Chains · *Streptococcus*
β-haem → *S. agalactiae, S. equi*
Bile-esculin + → *Enterococcus*

RODS — SPORES?

Spore-forming
Aerobe → *Bacillus*
Anaerobe → *Clostridium*
(see sheet 04)

Non-sporing
Pleomorphic, granules → *Coryne.*
Tumbling motility → *Listeria*
H_2S in TSI, non-motile → *Erysipel*

Staphylococcus SYSTEMATIC

- **Morphology:** Gram +ve cocci ~1 µm in **grape-like clusters**; non-motile, non-sporing, catalase +ve, facultative anaerobe.
- **Culture:** Golden-yellow colonies on nutrient agar; **β-haemolysis** on blood agar; grows in **7.5–10% NaCl** (Mannitol Salt Agar — mannitol fermented, medium turns yellow); Baird-Parker = black shiny colonies with clear halo.
- **Key test: Coagulase** — slide test detects bound clumping factor, tube test detects free coagulase. Coagulase +ve = pathogenic *S. aureus*.
- **Other tests:** DNase +, phosphatase +, protein A present, ferments mannitol anaerobically.
- **Virulence:** Protein A (binds Fc of IgG), coagulase, hyaluronidase, staphylokinase, lipase, α/β/γ/δ haemolysins, **Panton-Valentine leucocidin**, exfoliative toxin, **TSST-1 & enterotoxins A–E (superantigens, heat-stable)**.
- **Veterinary diseases:** bovine mastitis (chief contagious cause, hard fibrotic quarter, gangrenous mastitis in sheep = "blue bag"), tick pyaemia in lambs, **bumblefoot** and synovitis in poultry, botryomycosis, pyoderma (*S. pseudintermedius* in dogs), exudative epidermitis/greasy pig disease (*S. hyicus*).
- **Public health:** staphylococcal food poisoning — 2–6 h, vomiting, from enterotoxin in milk products; **MRSA** (*mecA*).

Differentiate: *Staphylococcus* catalase +, clusters, coagulase ± · *Micrococcus* catalase +, oxidative, modified oxidase + · *Streptococcus* catalase –, chains.

Streptococcus & *Enterococcus* SYSTEMATIC

- **Morphology:** Gram +ve cocci in **chains/pairs**, non-motile, **catalase –ve**, capnophilic, need enriched media.
- **Haemolysis classification: α** = partial, green (*S. pneumoniae*, viridans) · **β** = complete, clear (*S. agalactiae*, *S. equi*) · **γ** = none (*Enterococcus*).
- **Lancefield grouping** — precipitation against group-specific **C carbohydrate** of the cell wall (Rebecca Lancefield).

Group	Species	Disease
A	*S. pyogenes*	Human; strangles-like in primates
B	*S. agalactiae*	**Contagious bovine mastitis** — lives in udder, spread at milking
C	*S. equi* subsp. *equi*	**Strangles** — abscessed submandibular & retropharyngeal nodes
C	*S. dysgalactiae*	Environmental/summer mastitis, polyarthritis in lambs
D	*Enterococcus faecalis/faecium*	Enteric, UTI, VRE; bile-esculin +, 6.5% NaCl +
E	*S. porcinus*	Jowl abscess in pigs
—	*S. uberis, S. suis*	Environmental mastitis; meningitis in piglets (**zoonotic**)

Tests CAMP test + = *S. agalactiae* (arrowhead haemolysis with *S. aureus*) · Bacitracin sensitive = group A · Optochin sensitive & bile soluble = *S. pneumoniae* · **Hippurate hydrolysis + = *S. agalactiae*.**

Corynebacterium, *Trueperella*, *Rhodococcus* SYSTEMATIC

- **Corynebacterium** — Gram +ve, **pleomorphic club-shaped** rods in **Chinese-letter / palisade / V-L** arrangement; **metachromatic (Babes-Ernst) granules** shown by Albert's / Neisser's stain; non-motile, non-sporing, catalase +.
- *C. pseudotuberculosis* → **caseous lymphadenitis** of sheep & goats (onion-ring greenish caseous nodes), ulcerative lymphangitis in horses. Toxin: **phospholipase D**. Reverse CAMP with *R. equi*.
- *C. renale* group → **bovine pyelonephritis & cystitis**; pizzle rot (ulcerative posthitis) in wethers; urease strongly +.
- *C. diphtheriae* → diphtheria in man; toxin is phage(β)-coded; Loeffler's serum slope, potassium tellurite (black colonies); **Elek's gel precipitation test** for toxigenicity.
- ***Trueperella pyogenes*** (ex-*Arcanobacterium/Corynebacterium pyogenes*) — the classic **pus-forming** organism of cattle: summer mastitis, abscess, endometritis, pneumonia, arthritis. Small β-haemolytic colonies at 48 h, CAMP +.
- ***Rhodococcus equi*** — **partially acid-fast** coccobacillus, salmon-pink mucoid colonies, urease +. **Suppurative bronchopneumonia with abscesses in foals 1–4 months**; ulcerative enteritis. Virulence plasmid **VapA**; survives inside macrophages.

Listeria vs *Erysipelothrix* TABLE

Feature	*L. monocytogenes*	*E. rhusiopathiae*
Motility	**Tumbling at 25 °C**, umbrella growth in semisolid	Non-motile
Catalase	Positive	**Negative**
H_2S in TSI	Negative	**Positive**
Haemolysis	Narrow β (listeriolysin O); **CAMP +**	α on blood agar
Growth trick	**Cold enrichment 4 °C** — psychrotroph	"Test-tube brush" in gelatin stab
Disease	Circling disease/encephalitis (ruminants), abortion, septicaemia in monogastrics; **silage disease**	**Swine erysipelas** — rhomboid "diamond skin"; arthritis, endocarditis; ovine polyarthritis; turkey septicaemia
Zoonosis	Listeriosis — abortion, meningitis (immunocompromised)	**Erysipeloid** — fish-handler's / whale-finger
Immunity	Classic model of **cell-mediated immunity**; intracellular, ActA-driven actin tails	Neuraminidase, hyaluronidase; bacterin/live vaccine

Trap: *Listeria* is a facultative **intracellular** pathogen — escapes phagosome using **listeriolysin O** and moves cell to cell with actin tails, so antibody is nearly useless and **T-cell immunity** is the answer.

Mastitis pathogens — the table examiners love TABLE

Type	Organism	Reservoir & spread	Signature
Contagious	*Staph. aureus*	Infected udder → milking	Chronic, fibrosis, gangrene
	Strep. agalactiae	Obligate udder parasite	High SCC, eradicable
	Mycoplasma bovis	Udder, respiratory	Multiple quarters, no fever response to antibiotics

Sheet 03 · 60-second self-test RECALL

Prompt	Answer
Single best test to separate Staph from Strep	Catalase
Marker of pathogenic *S. aureus*	Coagulase
Selective medium for staphylococci	Mannitol salt / Baird-Parker
Lancefield antigen	Cell wall C carbohydrate
CAMP test positive species	*S. agalactiae, L. monocytogenes*
Cause of strangles	*S. equi* subsp. *equi*
Caseous lymphadenitis toxin	Phospholipase D
Foal pneumonia with abscesses	*Rhodococcus equi* (VapA)
Cold enrichment organism	*Listeria monocytogenes*
Diamond skin disease	*Erysipelothrix rhusiopathiae*

Type	Organism	Reservoir & spread	Signature
Environmental	*E. coli*, *Klebsiella*	Bedding, faeces	Acute toxic (endotoxic) mastitis
	Strep. uberis/*dysgalactiae*	Environment, skin	Clinical, dry-period onset
	T. pyogenes + *F. necrophorum*	Flies (*Hydrotaea*)	**Summer mastitis** — foul-smelling pus, quarter lost

Screening **CMT / SFMT** (California & Surf Field Mastitis Test) — detects DNA of somatic cells; **SCC > 2 lakh/ml** = subclinical mastitis. Confirm by culture + ABST before dry-cow therapy.

04 Spore-Formers, Acid-Fast & Filamentous Bacteria

t17–t20

Bacillus anthracis — the complete answer

SYSTEMATIC

- **Morphology:** large Gram +ve rod 1–1.5 × 3–8 µm with **square/truncate ends**; in culture long chains = **bamboo-stick / boxcar**; capsule of **poly-D-glutamic acid** (pXO2); **central, ellipsoidal, non-bulging spore**.
- **Golden rule:** spores form **only outside the body in free oxygen** — never in a living host or unopened carcass. **Never necropsy a suspect case**; spores survive in soil 50+ years ("incubator areas").
- **Culture:** non-haemolytic on sheep blood agar (*B. cereus* is β-haemolytic); **Medusa-head** colony edge; **inverted fir-tree** in gelatin stab; **string-of-pearls** with 0.5 IU/ml penicillin; grows on bicarbonate agar + 5% CO_2 to make capsule.
- **Toxin (pXO1, tripartite, A-B type): PA** (binds CMG2/TEM8, forms heptamer pore) + **EF** (calmodulin-dependent adenylate cyclase → oedema) + **LF** (Zn-metalloprotease cleaving MAPKK → macrophage death, shock).
- **Virulence needs BOTH plasmids.** Sterne 34F_2 vaccine = pXO2$^-$ (non-capsulated, toxigenic, live spore) — the vaccine used in India. Pasteur's was heat-attenuated.
- **Diagnosis:** peripheral blood/ear-vein smear with **McFadyean's polychrome methylene blue** — blue rods with pink capsule; **Ascoli thermoprecipitation test** on decomposed tissue/hides; PCR for *pagA*, *capB*; culture; animal inoculation in guinea pig.
- **Clinical:** peracute sudden death in cattle/buffalo/sheep; **dark tarry unclotted blood from natural orifices, no rigor mortis, enlarged friable "blackberry-jam" spleen**. Man: cutaneous **malignant pustule**, woolsorter's disease, intestinal.
- **Control:** do not open carcass; burn or bury 6 ft deep with quicklime; 10% formaldehyde/5% NaOH; annual Sterne vaccination in endemic belt; notifiable.

***B. anthracis* vs *B. cereus*:** non-motile / motile · non-haemolytic / β-haemolytic · capsulated / non-capsulated · penicillin-sensitive / resistant · susceptible to gamma phage & string-of-pearls / not. *B. cereus* = food poisoning (emetic & diarrhoeal) and bovine mastitis.

Mycobacterium

SYSTEMATIC

- **Acid-fastness** is due to **mycolic acid** in a lipid-rich (up to 60%) wall + peptidoglycolipid (**wax D**), demonstrated by **Ziehl-Neelsen**. Slender, beaded, non-motile, non-sporing, obligate aerobe.
- **Media: Lowenstein-Jensen** (egg, malachite green, glycerol) · Stonebrink (pyruvate) for *M. bovis* · Middlebrook 7H10/7H11 · **Dorset egg**, Herrold's egg-yolk with mycobactin J for *M. paratuberculosis*.
- **Growth:** slow — *M. tuberculosis* 2–4 wks (eugonic, rough buff tough), *M. bovis* 6–8 wks (**dysgonic**, needs pyruvate, glycerol inhibits), *M. avium* 2–3 wks, *M. paratuberculosis* 8–16 wks (**mycobactin-dependent** — the defining test).
- **Runyon groups** (atypical/NTM): I photochromogen · II scotochromogen · III non-chromogen · IV rapid grower (*M. fortuitum*).
- **Bovine TB (*M. bovis*):** tubercle = caseation + calcification; miliary or organ TB; **zoonotic via raw milk**. Diagnosis: **single intradermal / single intradermal comparative cervical tuberculin test** (avian vs bovine PPD, read at 72 h), **IFN-γ (Bovigam) assay**, ZN of milk/tissue, culture, PCR. Test-and-slaughter policy.
- **Johne's disease (*M. avium* subsp. *paratuberculosis*):** chronic granulomatous enteritis of cattle & buffalo; incubation 2–5 yrs; **corrugated, thickened "brain-like" ileum**, no ulcer; intermittent pipe-stream diarrhoea, emaciation, submandibular oedema, normal appetite. Diagnosis: ZN of rectal pinch/faeces (clumps of AFB), **johnin test**, AGID, ELISA, faecal culture, IS*900* PCR.
- **Avian TB (*M. avium*):** nodules in liver, spleen, intestine **without calcification**; positive avian tuberculin.
- **Virulence:** cord factor (trehalose-6,6′-dimycolate), sulpholipids, LAM, catalase-peroxidase; blocks phagosome-lysosome fusion → survives in macrophage → **Type IV hypersensitivity & granuloma**.

Clostridium — split by what the toxin does

TABLE

Group	Species	Disease & key toxin
Neurotoxic	*C. tetani*	**Tetanus** — drumstick/tennis-racquet terminal round spore, swarming, **tetanospasmin** blocks glycine/GABA at Renshaw cells → spastic paralysis, lockjaw, sawhorse stance, third-eyelid prolapse. Toxoid (ATT) + antitoxin
	C. botulinum	**Botulism** — types A–G; C & D in animals; **preformed toxin** in spoiled silage/carcass; flaccid paralysis, "limberneck" in birds. Type C = "lamziekte" with osteophagia in P-deficient cattle
Histotoxic	*C. chauvoei*	**Black quarter/blackleg** — endogenous, crepitant emphysematous myositis of heavy muscle in young cattle; rancid butter smell
	C. septicum / *novyi* / *sordellii*	Malignant oedema, braxy (abomasitis), **black disease** (*C. novyi* B with *Fasciola* migration), big head of rams, gas gangrene
Entero-toxigenic	*C. perfringens* A–E	**A** gas gangrene, yellow lamb disease · **B** lamb dysentery · **C** struck, necrotic enteritis · **D pulpy kidney/enterotoxaemia (ε-toxin)**, glycosuria · **E** enteritis. Type A also = **necrotic enteritis in broilers**
	C. difficile	Toxins A & B; colitis in horses/pigs after antibiotics

Toxin typing **α**=lecithinase (all types, **Nagler reaction** on egg-yolk agar, half-plate antitoxin) · **β**=B,C (trypsin-labile — why lambs on colostrum are hit) · **ε**=B,D · **ι**=E. Confirm by mouse neutralisation / multiplex PCR.

Blackleg vs malignant oedema: BQ = endogenous, no wound, heavy muscles, *C. chauvoei*. MO = wound-associated, subcutaneous, *C. septicum* chiefly, oedema > gas.

Actinomyces , *Nocardia* , *Streptomyces* , *Dermatophilus*

TABLE

Genus	O_2 & acid-fast	Disease
Actinomyces bovis	Anaerobe/capnophile · **Not** acid-fast	**Lumpy jaw** — pyogranulomatous osteomyelitis of mandible/maxilla in cattle; **sulphur granules** in pus (club colonies)
Nocardia asteroides	Strict aerobe · **Partially acid-fast** (modified ZN)	Bovine nocardial mastitis, canine/feline pyothorax, granulomatous pneumonia; branching filaments, soil saprophyte
Streptomyces	Aerobe · not acid-fast	Mostly soil saprophyte; **source of most antibiotics** (streptomycin, tetracycline); rare mycetoma
Dermatophilus congolensis	Facultative anaerobe · not acid-fast	**Dermatophilosis / rain scald / strawberry footrot**; branching filaments dividing in two planes → **"railroad-track" parallel rows of cocci (zoospores)**; motile flagellated zoospores activated by moisture

Actinobacillosis vs Actinomycosis: *Actinobacillus lignieresii* = **wooden tongue**, Gram −ve, soft tissue. *Actinomyces bovis* = **lumpy jaw**, Gram +ve, bone. Both give sulphur granules — the Gram reaction of the granule is the differentiator.

Sterilisation & disinfection — one-glance table

TABLE

Method	Condition	Use
Hot air oven	160 °C / 2 h	Glass, oil, powder
Autoclave	121 °C, 15 psi, 15 min	Media, instruments; **gold standard**
Tyndallisation	100 °C / 20 min × 3 d	Sugars, gelatin
Inspissation	80–85 °C / 30 min × 3 d	LJ, Loeffler serum
Pasteurisation	LTLT 63 °C/30 min · HTST 72 °C/15 s	Milk — kills *M. bovis*, *Coxiella*
Filtration	0.22 µm membrane	Sera, antibiotics (not viruses)
Radiation	UV 260 nm (surface) · gamma (ionising)	Cabinets · plasticware
Chemical	Ethylene oxide, glutaraldehyde, formalin, phenol, hypochlorite, quaternary ammonium	Heat-labile items, premises

Sheet 04 · 60-second self-test

RECALL

Prompt	Answer
Anthrax capsule chemistry	Poly-D-glutamic acid (pXO2)
Anthrax toxin components	PA + EF + LF
Anthrax field test on hides	Ascoli thermoprecipitation
Anthrax vaccine strain	Sterne 34F_2 (pXO2$^-$)
Chemical basis of acid-fastness	Mycolic acid
Growth factor for *M. paratuberculosis*	Mycobactin J
PCR target for Johne's	IS*900*
Terminal round spore, drumstick	*C. tetani*
Nagler reaction detects	*C. perfringens* α-toxin (lecithinase)
Pulpy kidney toxin	ε-toxin, *C. perfringens* type D
Sulphur granules, Gram +ve, bone	*Actinomyces bovis*
Railroad-track zoospores	*Dermatophilus congolensis*

Controls Autoclave → *Geobacillus stearothermophilus* spores · Hot air oven → *Bacillus atrophaeus* (*subtilis*) spores · Ethylene oxide → *B. atrophaeus*. Browne's tube & autoclave tape = chemical indicators. **Rideal-Walker / Chick-Martin** = phenol coefficient.

05 Gram-Negative Rods — Enteric & Respiratory Block

t21–t28

Identifying a Gram-negative rod in six moves

MECHANISM

Gram −ve rod on MacConkey

OXIDASE

Oxidase −ve · Enterobacteriaceae · glucose fermenter · nitrate → nitrite

LACTOSE

- **Lactose FERMENTER**
  Pink on MacConkey
  E. coli (IMViC ++−−, green sheen EMB)
  Klebsiella (mucoid, non-motile, VP+)
- **NON-fermenter**
  Pale colonies
  H_2S+, motile → *Salmonella*
  Swarms, urease++ → *Proteus*; *Yersinia* 22 °C motile

Oxidase +ve · non-enteric

GROWTH ON MacCONKEY

- **Grows**
  Pigment, grape odour → *Pseudomonas*
  Safety-pin, glanders → *Burkholderia*
  Curved, TCBS → *Vibrio*
- **Poor / no growth**
  Bipolar coccobacillus → *Pasteurella*
  X & V factors → *Haemophilus*
  CO_2-dependent → *Brucella*; seagull

E. coli, *Klebsiella* & the pathotypes

SYSTEMATIC

- **Family traits:** Gram −ve rods, **oxidase −ve**, ferment glucose, reduce nitrate, catalase +, facultative anaerobe. Antigens **O (somatic, LPS, heat-stable) · H (flagellar, protein, heat-labile) · K (capsular) · Vi (*S. Typhi*, *Citrobacter*)**.
- ***E. coli***: motile, lactose +, **IMViC = + + − −**, metallic green sheen on EMB, MR +, VP −, citrate −, indole +.

Pathotype	Mechanism	Veterinary disease
ETEC	Fimbriae **K88/F4, K99/F5, 987P, F41** + LT/ST enterotoxin	Neonatal calf/piglet/lamb diarrhoea (secretory, <1 wk)
EPEC	Intimin (*eae*) → **attaching-effacing** lesion	Diarrhoea in rabbits, dogs, calves
STEC/VTEC	Shiga toxin Stx1/Stx2	**Oedema disease** of weaned pigs (Stx2e); **O157:H7** HUS in man — cattle are the reservoir
Septicaemic	Capsule, aerobactin, serum resistance	Colisepticaemia in calves (failure of passive transfer), **colibacillosis in poultry** — pericarditis, perihepatitis, airsacculitis
UPEC / mastitis	P fimbriae; endotoxin	Cystitis, pyometra; **acute toxic coliform mastitis**

Klebsiella pneumoniae**: non-motile**, prominent polysaccharide capsule, mucoid stringy colonies, **VP +, citrate +, urease weakly +, indole −** (mirror image of *E. coli*). Environmental mastitis from sawdust bedding, metritis in mares, pneumonia in foals, CAUTI. Multi-drug resistant (ESBL, carbapenemase).

Salmonella, *Yersinia*, *Proteus*

SYSTEMATIC

- ***Salmonella***: motile (except ***S.*** **Gallinarum & Pullorum**), non-lactose fermenter, **H_2S +**, urease −, indole −, citrate +. TSI = **alkaline slant / acid butt with blackening**. Media: selenite F / tetrathionate broth → XLD, BSA (Wilson-Blair, **black colonies**), MacConkey pale.
- **Kauffmann-White scheme** classifies >2600 serovars on O, H (phase 1 & 2) and Vi antigens.
- **Diseases:** calf paratyphoid (*S.* Dublin, Typhimurium) — septicaemia, enteritis, abortion; **abortion in mares (*S.* Abortusequi)** and ewes (*S.* Abortusovis); **fowl typhoid (*S.* Gallinarum)** and **pullorum disease/BWD (*S.* Pullorum, egg-transmitted)** — detected by **rapid serum/whole-blood plate agglutination**; swine salmonellosis (*S.* Choleraesuis). Food-borne zoonosis via eggs and meat.
- **Virulence:** SPI-1 & SPI-2 pathogenicity islands (T3SS), Vi capsule, endotoxin, survival in macrophages.
- ***Yersinia*****: bipolar staining, motile at 22 °C but non-motile at 37 °C**, urease +. *Y. pestis* — plague, safety-pin appearance, flea-borne, **rodent reservoir**, Fraction 1 capsule, V & W antigens. *Y. enterocolitica* — cold enrichment, enterocolitis, pigs are the reservoir. *Y. pseudotuberculosis* — caseous nodules in rodents, birds, sheep.
- ***Proteus*****: swarming** on blood agar (inhibited by 5% agar / CLED), **urease strongly +** (→ struvite urolith & alkaline urine), H_2S +, PDA +, fishy odour. UTI, otitis externa, wound infection. **Weil-Felix** — OX-19/OX-2/OX-K strains cross-react with rickettsial antibodies.

Pasteurella · *Mannheimia* · *Actinobacillus* · *Haemophilus*

TABLE

Organism	Disease & key point
P. multocida **B:2 / E:2**	**Haemorrhagic septicaemia** of cattle & buffalo — monsoon, sudden death, hot painful submandibular swelling, petechiae on serosa. Bipolar Gram −ve coccobacillus (Leishman/methylene blue), capsulated, non-motile, indole +, no growth on MacConkey. **Vaccine: alum-precipitated / oil-adjuvant HS bacterin**
P. multocida **A:1**	**Fowl cholera** — greenish diarrhoea, wattle abscess, petechiae in heart fat; also progressive atrophic rhinitis in pigs (with *Bordetella*) via dermonecrotic toxin
Mannheimia haemolytica **A1**	**Shipping fever / enzootic pneumonia** of cattle & sheep; **leukotoxin (RTX)** kills ruminant leucocytes; β-haemolytic, grows on MacConkey, indole −
Actinobacillus lignieresii	**Wooden tongue** in cattle — soft-tissue granuloma, sulphur granules, Gram −ve rosettes
A. equuli **/** ***A. pleuropneumoniae***	Sleepy foal disease (joint-ill, nephritis) · **Porcine pleuropneumonia** — Apx toxins, NAD-dependent, CAMP +
Haemophilus	Needs **X (haemin) and/or V (NAD)** factors → **satellitism** around *S. aureus* on blood agar. *H. parasuis* = **Glasser's disease** (polyserositis, arthritis, meningitis); *Avibacterium paragallinarum* = **infectious coryza** in poultry

Brucella

SYSTEMATIC

- Gram −ve, **tiny coccobacilli, non-motile, non-sporing, non-capsulated**, aerobic, **catalase & oxidase +, urease +, partially acid-fast (modified ZN → red on blue)**.
- **Species & host:** *B. abortus* (cattle) · *B. melitensis* (goat/sheep — most pathogenic for man) · *B. suis* (pig) · *B. ovis* (ram epididymitis) · *B. canis* (dog) · *B. neotomae*, *B. ceti*, *B. pinnipedialis*.
- **Growth aids:** *B. abortus* biovar 1 needs **5–10% CO_2**; differentiate species by CO_2 need, H_2S, urease speed, dye inhibition (**thionine & basic fuchsin**), phage Tbilisi lysis, A/M monospecific sera.
- **Pathogenesis:** facultative intracellular in macrophages & trophoblasts; **erythritol** in the placenta of cattle, sheep, goat, pig favours growth → **abortion in the last trimester**, retained placenta, orchitis/epididymitis, hygroma. Man: **undulant/Malta fever** — a major lab-acquired infection.
- **Tests:** RBPT (screening) · **STAT** · MRT (milk ring, herd screening) · 2-ME/Rivanol (IgG, chronic) · CFT (OIE international trade standard) · **i-ELISA/c-ELISA** · FPA · AGID for *B. ovis* · PCR (*bcsp31*, *IS711*) · culture on Brucella selective agar.
- **Vaccine:** ***B. abortus*** **S19** — live, 4–8 month heifer calves, single dose, gives lifelong immunity

Pseudomonas, *Vibrio*, *Campylobacter*, *Bordetella*, *Moraxella*

RAPID-FIRE

P. aeruginosa	Oxidase +, non-fermenter, **pyocyanin (blue) + pyoverdin (green)**, sweet grape/corn-tortilla odour, grows at 42 °C. Otitis externa in dogs, mastitis from contaminated water, burn/wound sepsis, **notoriously multi-drug resistant**; ExoA toxin
Burkholderia mallei	**Glanders** in equines — farcy nodules along lymphatics, nasal ulcers; non-motile; **Mallein test**, CFT; **notifiable & zoonotic**; no treatment, destruction policy
B. pseudomallei	Melioidosis — soil/water saprophyte, motile, safety-pin, caseous nodules
Vibrio cholerae	Curved comma rod, single polar flagellum, **darting motility**, grows at **pH 8.5–9.0**, TCBS **yellow** colonies (sucrose +), cholera toxin ↑cAMP; O1 (classical, El Tor) & O139
Campylobacter fetus	**Seagull-wing / S-shaped**, microaerophilic (5% O_2, 10% CO_2), 42 °C for *C. jejuni*. *C. fetus* subsp. *venerealis* = **bovine genital campylobacteriosis** — infertility, early embryonic death, repeat breeding; subsp. *fetus* = ovine abortion. *C. jejuni* = leading food-borne zoonotic enteritis, poultry reservoir, **Guillain-Barré** sequel
Bordetella bronchiseptica	Motile, oxidase & urease +. **Kennel cough** complex, atrophic rhinitis in pigs (dermonecrotic toxin + *P. multocida* toxigenic strain), bronchopneumonia. *B. avium* = turkey coryza
Moraxella bovis	**Infectious bovine keratoconjunctivitis — pink eye**; fimbriae + β-haemolytic cytotoxin; spread by face flies & UV light; Gram −ve diplobacillus

but antibody interferes with testing. **RB51** — rough, no smooth-LPS antibody, DIVA-compatible. **Rev-1** for sheep & goats.

Trap: Bull semen and blood are NOT reliable samples — use **aborted foetal stomach content, placental cotyledon, vaginal discharge, milk**. No treatment in animals; policy is test-and-slaughter + vaccination.

IMViC & biochemical shorthand

TABLE

Organism	Ind	MR	VP	Cit	Ure	H_2S	Mot	Lac	Ox
E. coli	+	+	–	–	–	–	+	+	–
Klebsiella	–	–	+	+	w+	–	–	+	–
Salmonella	–	+	–	+	–	+	+	–	–
Proteus	±	+	–	±	++	+	+	–	–
Yersinia ent.	±	+	–	–	+	–	22°C	–	–
Pseudomonas	–	–	–	+	±	–	+	–	+
P. multocida	+	–	–	–	–	–	–	–	+
M. haemolytica	–	–	–	–	–	–	–	+	+
Brucella	–	–	–	–	+	±	–	–	+

IMViC Indole · **M**ethyl red · **V**oges-Proskauer · **C**itrate. *E. coli* = **+ + – –**; *Klebsiella/Enterobacter* = **– – + +**. That single contrast answers most viva questions on Enterobacteriaceae.

Sheet 05 · 60-second self-test

RECALL

Prompt	Answer
Single test separating enterics from the rest	Oxidase (enterics –ve)
Green metallic sheen on EMB	*E. coli*
Fimbria of calf-scour ETEC	K99 / F5
Oedema disease toxin	Stx2e (STEC)
Salmonella serotyping scheme	Kauffmann-White
Non-motile salmonellae	*S.* Gallinarum & Pullorum
Swarming, urease ++	*Proteus*
Weil-Felix strains	OX-19, OX-2, OX-K of *Proteus*
Placental growth factor for *Brucella*	Erythritol
DIVA brucella vaccine	RB51
Leukotoxin of shipping fever	*Mannheimia haemolytica* RTX
Mallein test diagnoses	Glanders (*B. mallei*)

06 Anaerobes, Spirochaetes, Mollicutes & Obligate Intracellulars

t29–t34

Gram-negative anaerobes

SYSTEMATIC

Organism	Veterinary significance
Fusobacterium necrophorum	Long filamentous Gram –ve rod. **Calf diphtheria** (necrotic laryngitis), **hepatic abscess of feedlot cattle** (with *T. pyogenes*), necrotic rumenitis after grain overload, **foot rot / interdigital necrobacillosis**, summer mastitis, thrush. Leukotoxin + endotoxin
Dichelobacter nodosus	Large Gram –ve rod with swollen ends. **Essential (transmitting) agent of ovine foot rot**; *F. necrophorum* is the predisposing invader. Produces **keratinase/protease** that digests hoof horn. Vaccine = pilus-based; footbath 10% $ZnSO_4$ / 5% formalin
Bacteroides fragilis **group**	Bile-resistant, non-sporing anaerobe; normal gut flora turned opportunist — abdominal abscess, peritonitis, aspiration pneumonia. Intrinsically β-lactamase producing; metronidazole is the drug
Porphyromonas/Prevotella	Black-pigmented anaerobes; periodontal disease of dog & cat

Anaerobic culture Freshly prepared, pre-reduced media · Robertson's cooked meat · thioglycollate · McIntosh-Fildes jar or GasPak (H_2 + CO_2, palladium catalyst) · anaerobic glove box. Foul putrid odour, gas, growth **away** from the surface in a shake tube.

Leptospira & other spirochaetes

SYSTEMATIC

- ***Leptospira interrogans*** (pathogenic) vs *L. biflexa* (saprophytic). Thin, tightly coiled, **hooked ends (question-mark shape)**, motile by two axial filaments/endoflagella; **too thin for Gram stain** — use **dark-field microscopy**, silver impregnation (Levaditi/Fontana), FAT.
- **Culture: EMJH** or Korthof's/Fletcher's semisolid at 28–30 °C, incubate up to **6–8 weeks**; growth as a Dinger's ring.
- **Serovars & hosts:** Pomona & Hardjo (cattle, pigs) · Canicola & Icterohaemorrhagiae (dog, rodent reservoir) · Grippotyphosa · Bratislava (horses). **Rodents are maintenance hosts and shed lifelong in urine.**
- **Disease:** leptospiraemia → fever, haemoglobinuria ("red water"), jaundice, **abortion & agalactia (milk drop syndrome, flabby-bag)**, interstitial nephritis, recurrent uveitis (periodic ophthalmia) in horses. Man: Weil's disease — occupational zoonosis of paddy workers, sewer workers, veterinarians.
- **Diagnosis: MAT** (microscopic agglutination test — reference standard, titre ≥1:100 or 4-fold rise), dark-field of fresh urine, ELISA, PCR (*lipL32*, 16S). Treat with streptomycin/oxytetracycline; vaccines are **serovar-specific bacterins**.
- **Others:** *Borrelia burgdorferi* — Lyme disease, *Ixodes* tick, erythema migrans, lameness in dogs; *B. anserina* — avian spirochaetosis (fowl tick *Argas*). *Treponema pallidum* — syphilis (non-cultivable). ***Brachyspira hyodysenteriae* — swine dysentery**, mucohaemorrhagic colitis, strongly β-haemolytic anaerobe.

Mycoplasma & *Ureaplasma*

SYSTEMATIC

- **Class Mollicutes — smallest free-living, self-replicating organisms** (0.2–0.3 µm); **no cell wall** → highly pleomorphic, Gram-unstainable (use Giemsa), **inherently resistant to penicillin & all β-lactams and to lysozyme**; membrane contains **sterol** (needs serum in medium); passes 0.45 µm filters.
- **Culture:** PPLO agar with 20% serum + yeast extract, 5–10% CO_2, humid; **"fried-egg" colonies** at 3–10 days; **Dienes stain**. *Ureaplasma* = urease +, tiny colonies.
- **Diseases: *M. mycoides* subsp. *mycoides* SC → CBPP** (contagious bovine pleuropneumonia, marbled lung, notifiable) · ***M. capricolum* subsp. *capripneumoniae* → CCPP** · *M. agalactiae* → contagious agalactia of sheep & goats (mastitis, arthritis, keratoconjunctivitis) · *M. bovis* → mastitis, pneumonia, arthritis, otitis in calves · ***M. gallisepticum* → chronic respiratory disease (CRD) of poultry, egg-transmitted**, detected by rapid serum plate agglutination & HI · *M. synoviae* → infectious synovitis · *M. hyopneumoniae* → enzootic pneumonia of pigs.
- **Treatment:** tylosin, tiamulin, tetracyclines, fluoroquinolones — **never penicillins**.

L-forms vs *Mycoplasma*: L-forms are cell-wall-deficient variants of ordinary bacteria induced by penicillin/lysozyme and **can revert**; *Mycoplasma* never had a wall and never reverts. Both make fried-egg colonies — the reversion test separates them.

Rickettsiales & *Chlamydia*

TABLE

Agent	Vector / route	Disease
Anaplasma marginale	Ticks (*Rhipicephalus*, *Hyalomma*), fomites, needles	**Bovine anaplasmosis** — marginal inclusion in RBC on Giemsa, severe anaemia **without haemoglobinuria**, icterus
Ehrlichia canis	*Rhipicephalus sanguineus*	Canine monocytic ehrlichiosis — morulae in monocytes, thrombocytopenia, epistaxis ("tracker dog disease")
Coxiella burnetii	**Aerosol** from birth fluids; ticks; extremely resistant spore-like form	**Q fever** — abortion in small ruminants; **zoonosis, very low infective dose**; phase I/II antigenic variation; survives pasteurisation poorly-done
Neorickettsia risticii	Trematode in aquatic insects	Potomac horse fever — colitis
Rickettsia **spp.**	Ticks, lice, fleas, mites	Spotted fever & typhus groups; **Weil-Felix** heterophile test
Chlamydia abortus, *C. psittaci*	Inhalation, ingestion	**Enzootic abortion of ewes**; **psittacosis/ornithosis** in birds & man; *C. felis* conjunctivitis; *C. pecorum* polyarthritis

Chlamydial cycle **Elementary body** = small, rigid, **infectious, metabolically inert** (the "spore") → endocytosis → reorganises into **Reticulate body** = larger, fragile, **non-infectious but dividing** → back to EB → release. Inclusions stained by Giemsa, **Gimenez**, Machiavello. Both *Chlamydia* and *Rickettsia* are **obligate intracellular** — they cannot be grown on cell-free media; use eggs, cell culture or lab animals.

Emerging, re-emerging & transboundary bacteria

RAPID-FIRE

Definitions	**Emerging** = newly appeared in a population. **Re-emerging** = previously controlled, now rising. **Transboundary (TAD)** = highly contagious, crosses borders, major trade impact — WOAH-listed & notifiable
Drivers	Intensification, live-animal trade, climate & vector range shift, wildlife interface, antimicrobial misuse, urbanisation, ecosystem change
Emerging bacteria	MRSA (LA-MRSA CC398), ESBL/carbapenemase *E. coli*, *Streptococcus suis*, *Campylobacter jejuni*, *Yersinia enterocolitica*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Mycoplasma bovis*
Re-emerging in India	Anthrax (endemic belts of Odisha, Andhra, Karnataka, Tamil Nadu), brucellosis, bovine TB, glanders in equines, leptospirosis
Transboundary bacterial	CBPP, CCPP, HS, anthrax, glanders, contagious agalactia, *Salmonella* Gallinarum
Bioterror agents (Cat A)	*B. anthracis*, *Y. pestis*, *F. tularensis*, botulinum toxin
Control framework	**One Health**; WOAH/FAO/WHO tripartite; NADCP & NAP-AMR in India; disease-free zoning & compartmentalisation; risk analysis, surveillance, biosecurity, ring vaccination, stamping out

Two skeletons that cover half the long questions

SKELETON

A · "Write about *[any bacterium]*" — 8-heading frame

1. Classification & important species
2. Morphology & staining (with the one signature feature)
3. Cultural & biochemical characters (name the selective medium)
4. Antigenic structure / typing scheme
5. Virulence factors & pathogenesis
6. Host, disease, lesions & clinical signs
7. Laboratory diagnosis — sample, direct exam, culture, biochemical, serological, molecular
8. Treatment, immunity, vaccine & control (add the Indian field angle)

B · "Laboratory diagnosis of a bacterial disease" — 6-step frame

1. **Sample:** right tissue, right time, aseptic, with transport medium & cold chain
2. **Direct microscopy:** Gram / ZN / capsule / dark-field / FAT
3. **Isolation:** enrichment → selective & differential media → pure culture; note colony & haemolysis
4. **Identification:** biochemical panel, serotyping, phage typing, MALDI-TOF
5. **Serology:** agglutination, CFT, AGID, ELISA — say what a titre means
6. **Molecular:** PCR/real-time PCR/sequencing; and **ABST by Kirby-Bauer** for the therapeutic answer

Unit 1 · final 90-second sweep

RECALL

Smallest free-living organism	*Mycoplasma* — no cell wall, sterol membrane	**Fried-egg colony**	*Mycoplasma* / L-forms
Dark-field organism	*Leptospira*	**Leptospira reference test**	MAT
Leptospira medium	EMJH	**CBPP agent**	*M. mycoides* subsp. *mycoides* SC
CRD of poultry	*M. gallisepticum*	**Infectious form of chlamydia**	Elementary body
Q fever agent	*Coxiella burnetii*	**Marginal RBC inclusion**	*Anaplasma marginale*
Essential agent of ovine footrot	*Dichelobacter nodosus*	**Calf diphtheria & liver abscess**	*Fusobacterium necrophorum*
Swine dysentery	*Brachyspira hyodysenteriae*	**Penicillin-resistant by nature**	*Mycoplasma* (no wall)
Obligate intracellular pair	*Rickettsia* & *Chlamydia*	**Heterophile test for rickettsiae**	Weil-Felix

`34 / 34 syllabus topics` `Paper I · Unit 1` `Full pass ≈ 30 min` `Night-before pass: sheets 05-06 + all self-tests`